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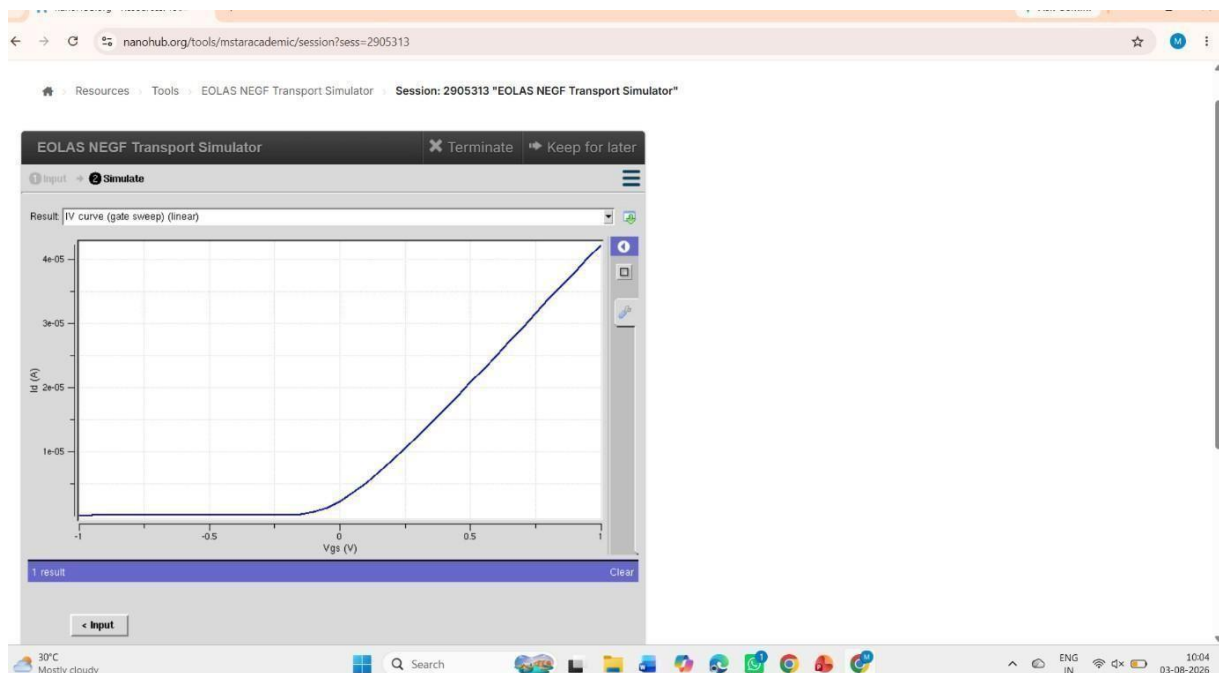
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Exp. 7. Simulation of VI Characteristics of FINFET, CNTFET and GFET using Nano Hub

Aim:

To Obtain the VI Characteristics of FinFET by varying the Gate Channel length and oxide thickness

(a) FINFET



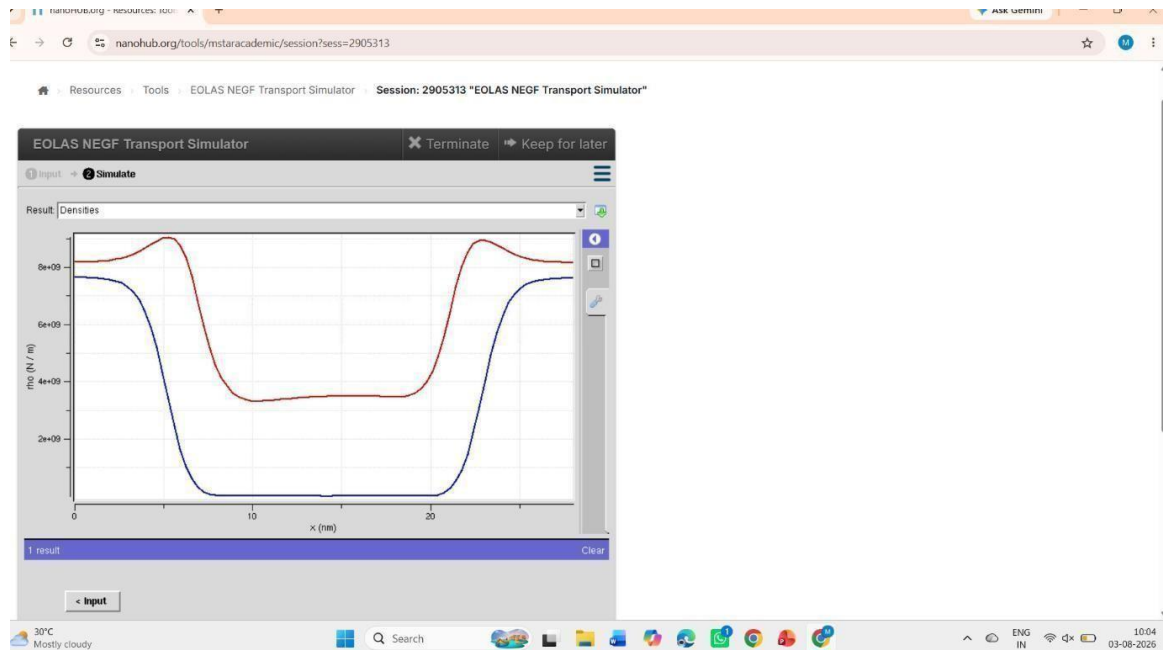
FinFET – Drain Current vs Gate Voltage: The graph represents the variation of drain current (I_d) with gate-source voltage (V_{gs}) for the FinFET. As the gate voltage increases, more charge carriers are attracted towards the channel, which increases the channel conductivity and consequently increases the drain current. The curve shows a significant rise in drain current with increasing gate voltage, indicating effective control of the channel by the gate. This characteristic demonstrates the switching behavior and strong gate control of the FinFET, which is important for high-performance and low-power electronic applications.

Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET . Analyze the performance for different Oxide layer thicknesses



FinFET – Energy-Resolved Current: The graph shows the variation of current with electron energy for different oxide-layer thicknesses in the FinFET. The current is mainly concentrated around the Fermi-level energy region and decreases as the energy increases. The different curves indicate that changes in oxide thickness can influence the carrier transport characteristics of the device. This analysis helps in understanding the relationship between energy distribution, oxide thickness, and electron transport in nanoscale FinFET devices.

Case Study 2: Simulate & obtain the electron density per sheet characteristics of a FinFET. Analyse its performance with different Gate region lengths



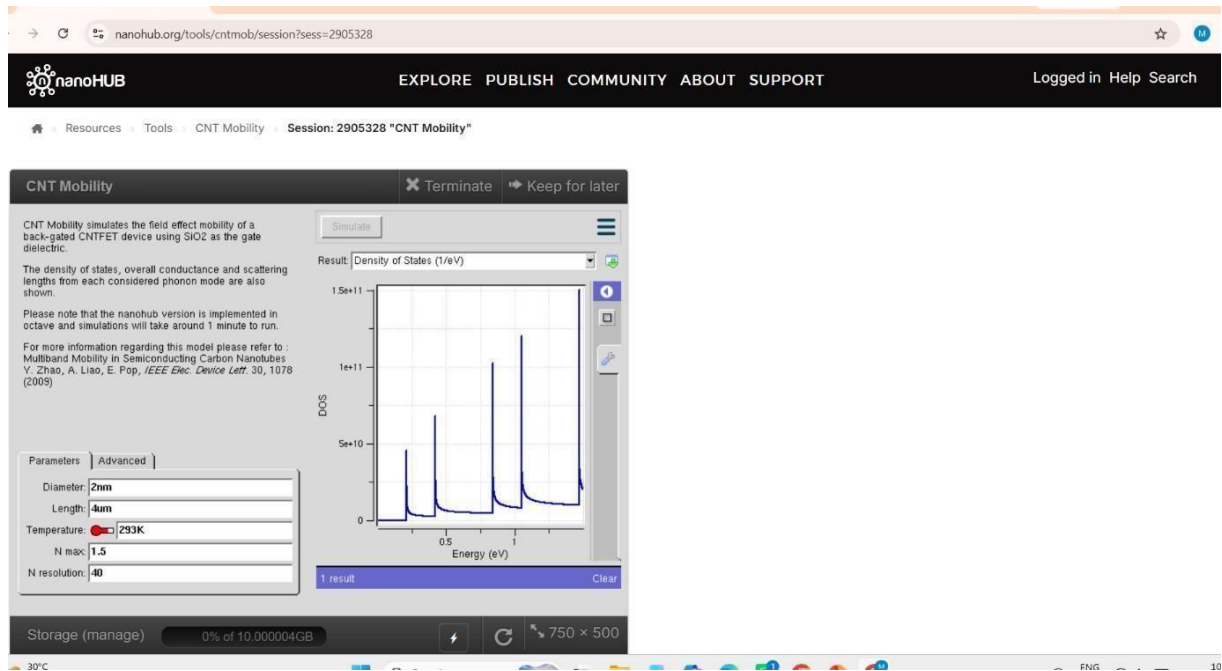
FinFET – Electron and Hole Density: The graph represents the distribution of electron and hole densities along the FinFET for different gate-region lengths. The carrier concentration changes from one region of the device to another, showing how electrons and holes are distributed within the semiconductor structure. The variation in carrier density provides information about carrier accumulation and transport near the source, channel, and drain regions. This graph is useful for studying the internal charge distribution and electrical behavior of the FinFET.

(b) CNTFET

Aim:

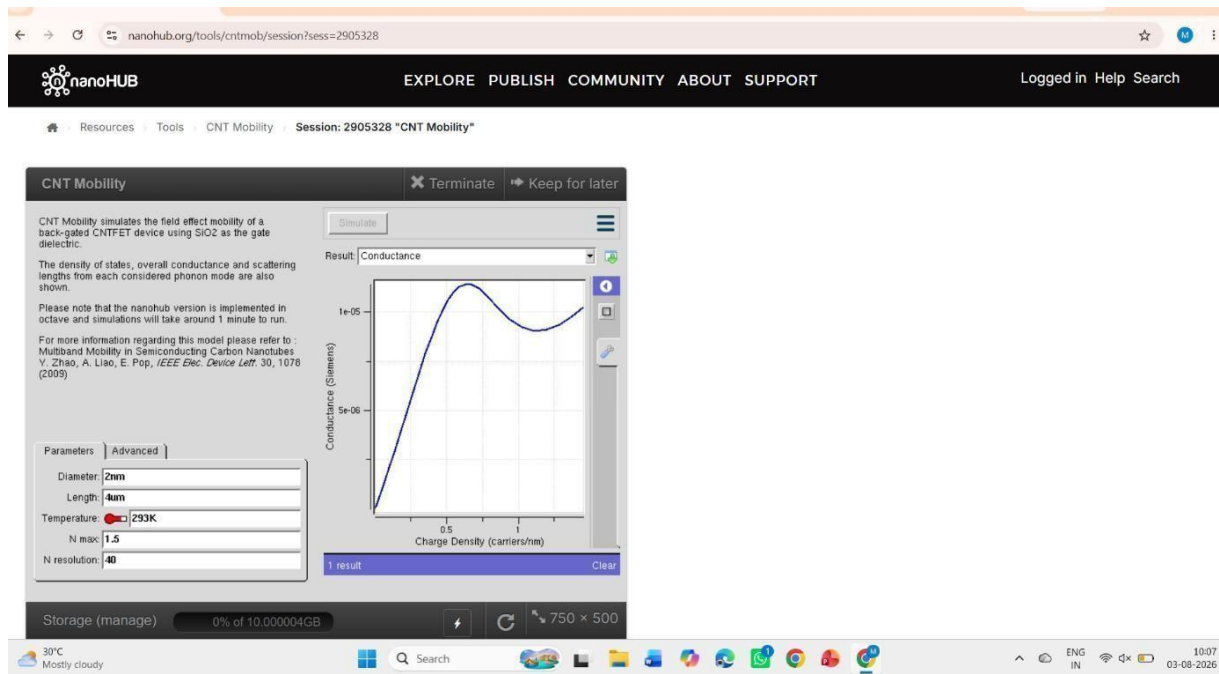
To obtain the density of states and conductance of CNTFET by varying the diameter and length.

Output:



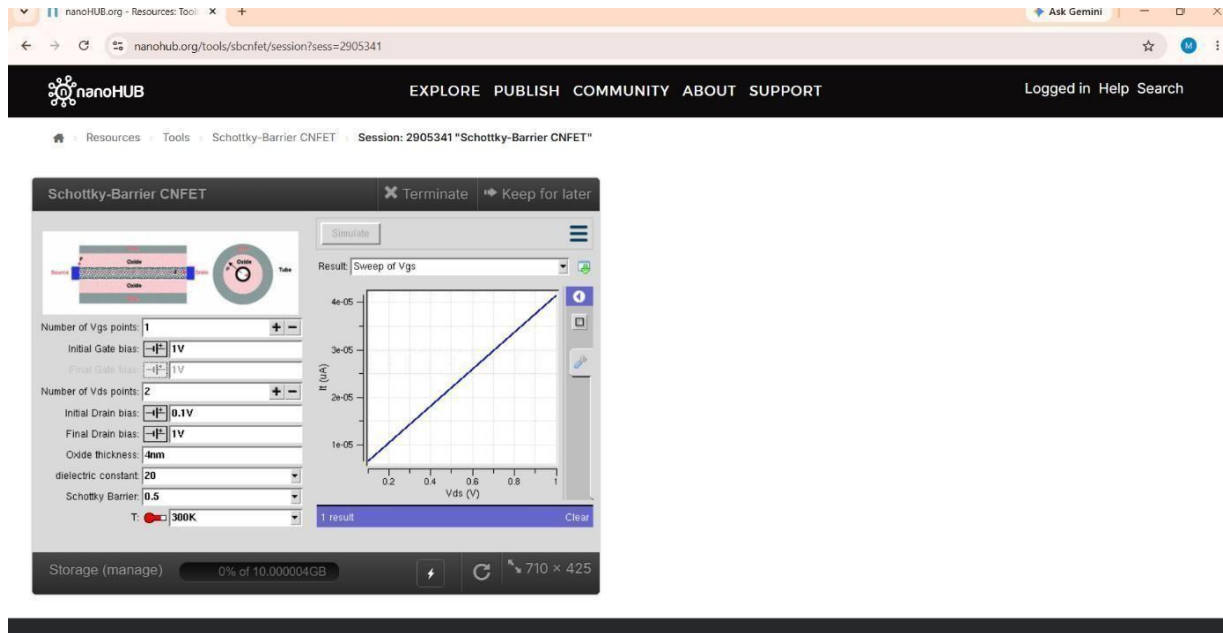
CNTFET – Density of States:

The graph represents the density of states (DOS) as a function of energy for the CNTFET. Several sharp peaks can be observed at specific energy levels, which indicate the availability of electronic states within the carbon nanotube. These peaks are a result of the quantized energy levels produced by the nanoscale dimensions of the CNT. The DOS characteristics are important for understanding the electronic structure, carrier availability, and quantum transport properties of CNT-based transistors.



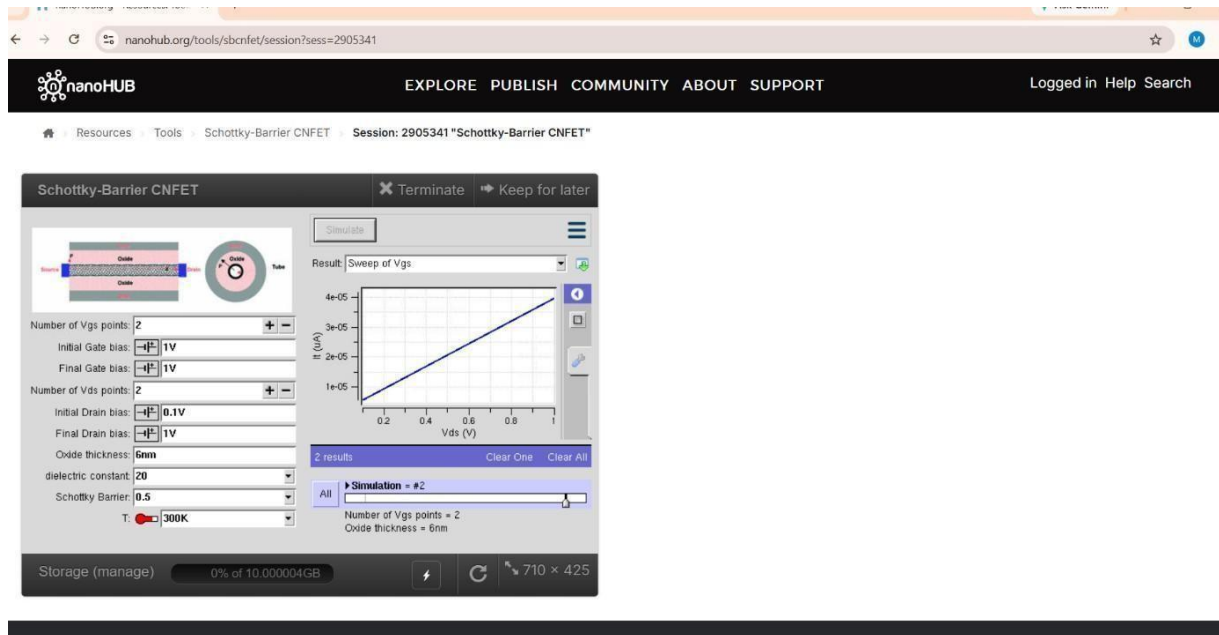
CNTFET – Conductance vs Carrier Density: The graph shows the relationship between conductance and carrier density in the CNTFET. As the carrier density increases, the conductance also increases because a greater number of charge carriers are available to participate in electrical conduction. The increasing conductance indicates improved transport through the carbon nanotube. Therefore, this graph demonstrates how carrier concentration influences the electrical performance and current-carrying capability of the CNTFET.

Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



CNTFET – VI Characteristics – Schottky Barrier & Oxide Thickness: The graph represents the drain current versus drain voltage characteristics of the CNTFET for different Schottky barrier and oxide-thickness conditions. As the drain voltage increases, the drain current also increases, showing the current conduction behavior of the device. The different curves indicate that the Schottky barrier and oxide thickness influence the current flowing through the CNT channel. This analysis is useful for understanding and optimizing contact properties, gate control, and current transport in CNTFETs.

Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter



CNTFET – VI Characteristics – Different CNT Diameters: The graph shows the drain current versus drain voltage characteristics for different CNT diameters and gate-bias conditions. The drain current increases with increasing drain voltage, while the different curves demonstrate the influence of CNT diameter and gate voltage on device performance. Changes in the nanotube diameter can modify the electronic and transport properties of the device. Hence, this graph helps in studying the effect of CNT dimensions and gate bias on the electrical characteristics of CNTFETs.

(c) GFET

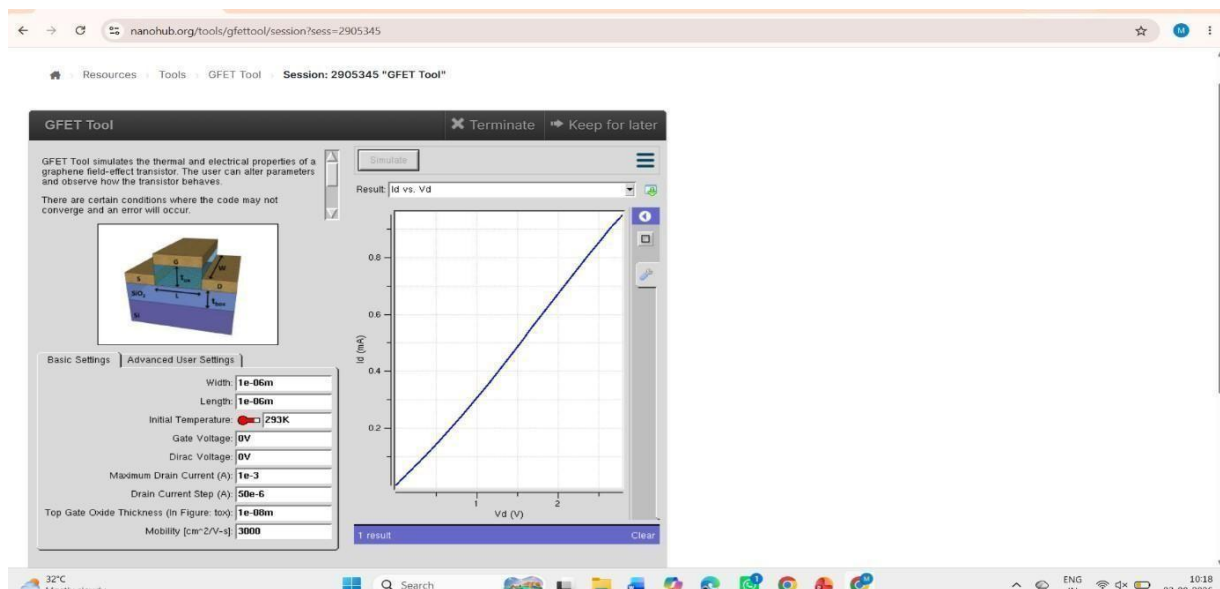
Aim:

To simulate the Voltage-Current (VI) characteristics of a Graphene Field Effect

Transistor (GFET) using appropriate simulation tools, analyze its performance under

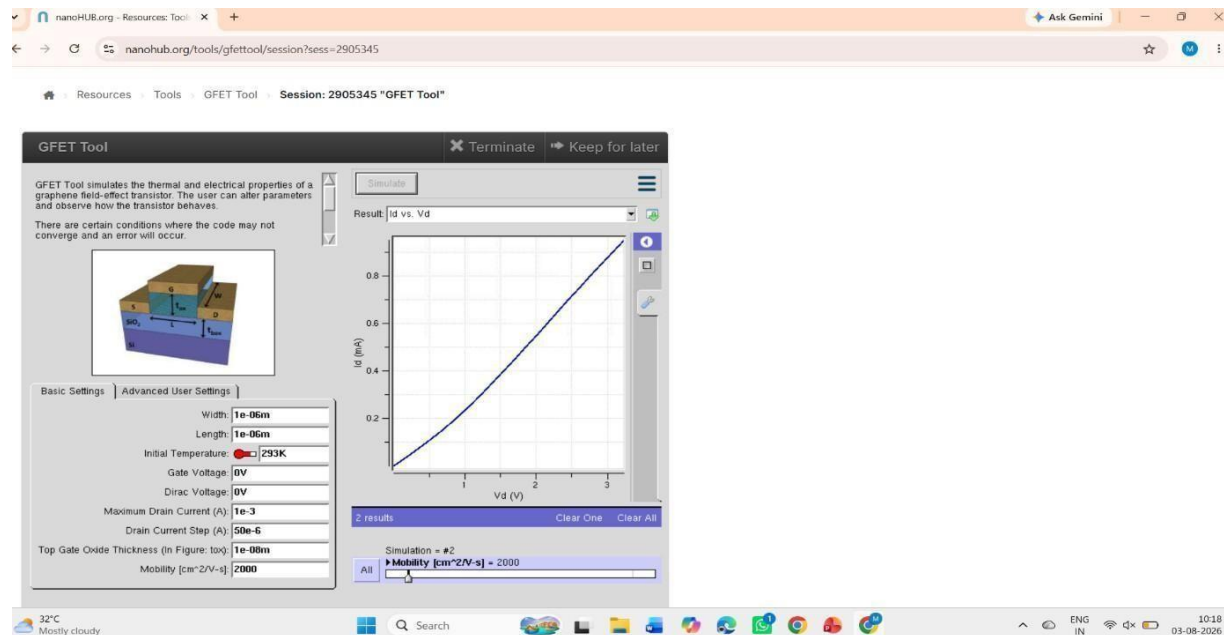
varying gate voltages and drain-source voltages, and understand its behavior as a high- mobility semiconductor device.

Case Study 1: How does varying the gate voltage (V_{gs}) affect the drain current (I_{ds}) in a GFET?



GFET – VI Characteristics: The graph presents the drain current versus drain voltage characteristics of the GFET for different gate voltages. The drain current increases with increasing drain voltage, showing good electrical conduction through the graphene channel. Different gate-voltage conditions produce different current levels, demonstrating the ability of the gate to control carrier transport. The graph therefore indicates the good conductivity, gate-controlled operation, and efficient carrier transport of graphene-based field-effect transistors.

Case Study 2: What factors determine the maximum drain current ($I_{ds,max}$) in a Graphene Field Effect Transistor (GFET)



GFET – Maximum Drain Current: The graph shows the drain current characteristics of the GFET under different gate-source voltage conditions. The curves indicate relatively high drain current for the applied gate voltages, demonstrating the strong carrier-transport capability of the graphene channel. The variation in current with gate voltage shows that the electrical behavior of the GFET can be controlled through the gate. This characteristic makes the GFET useful for studying high-speed operation, efficient carrier transport, and nanoscale electronic applications.

CONCLUSION: The simulated characteristics of FinFET, CNTFET, and GFET were compared based on their current conduction, carrier transport, and nanoscale electronic behavior. The FinFET shows strong gate control and effective switching, while the CNTFET exhibits quantized energy levels, increasing conductance with carrier density, and sensitivity to CNT diameter, Schottky barrier, and oxide thickness. The GFET demonstrates good electrical conduction and relatively high drain current through the graphene channel.

Among the three devices, GFET shows better carrier-transport characteristics based on the given simulation results, whereas FinFET is advantageous for gate-controlled switching and CNTFET is useful for studying quantum-scale transport. Overall, the experiment successfully demonstrates and compares the electrical characteristics of different nanoscale transistors and helps in understanding their suitability for advanced nanoelectronic and low-power applications.